# PARISHRAM 2.0

## CLASS 12TH 2026

# d & f-Block

CHEMISTRY | LECTURE-4

BY SANYA NAYAR MA'AM



# TOPICS TO BE COVERED

1. Properties of Transition Elements
2. Trends of Transition Elements




ROAR
Tiger hai tu Tiger! Roar

# Let's Recap

**When Tiger Ma'am says-**

**Pehle zara purana revise kar lete hai**



# Formation of Coloured Ions

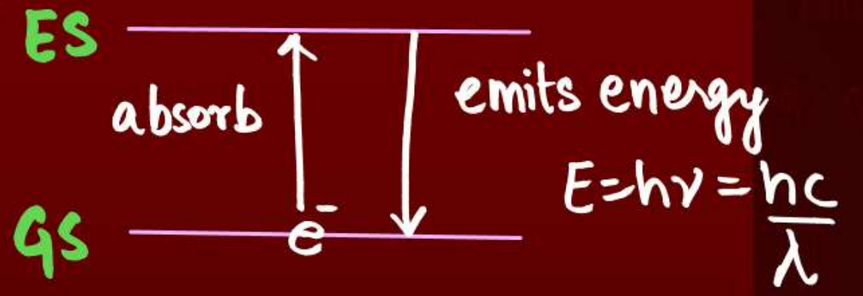
- Colour : Due to unpaired electrons → excitation (d-d transition)

$d^0$ and $d^{10}$ : **Colourless**

$d^0$ → (no d $e^-$s)

$d^{10}$ → (all $e^-$s are paired up)





# Formation of Coloured Ions

$x - 8 = -1$

$x = +7$

$MnO_4^-$ $Mn^{7+} : 3d^0$

Permanganate ion (Pink)

- Along with d-d transition, colour may be due to

i. **Charge transfer spectra**

e.g. $MnO_4^-$ is pink coloured due to charge transfer from ligand to metal (LMCT) $Fe_4[Fe(CN)_6]$ is blue coloured due to charge transfer from metal to metal.

ii. **Polarisation**

AgBr and AgI are coloured due to polarization.

Pale Yellow → Yellow

$Br^-$ $I^-$ ↑

Fajan's Rule: Smaller the size of cation, larger the size of anion, Greater is the covalent character

# Question

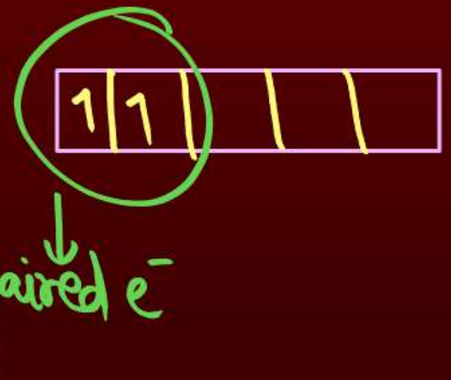
Which of the following ions is coloured in solution?

i) $Zn^{2+}$ : $3d^{10}$ ✗

ii) $Ti^{4+}$ : $3d^{0}$

iii) $Cu^{+}$ : $3d^{10}$

iv) $V^{3+}$ : $3d^{2}$ ✓



# Alloy Formation

Homogeneous mixture of 2 or more metals or a metal & a non-metal

- Due to almost similar size of metal ion (~15%), high ionic charges and availability of d-orbitals.

  e.g. Steel, brass, stainless steel, gold jewellery etc.

  (Fe + C) (Cu + Zn) → (Fe, Cr, Ni & traces of C)

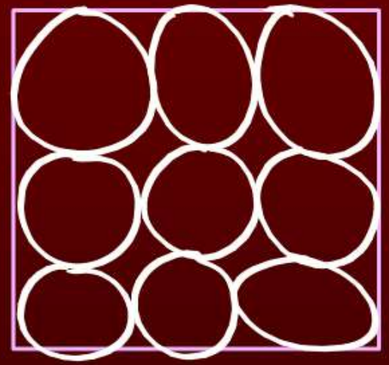
  ↳ Au + Cu/Ag

# Formation of Interstitial Compound

Small-sized atoms of H, B, C, N etc. can easily occupy positions in the voids present in the crystal lattices of transition metals.

- Usually non-stoichiometric and are neither typically ionic nor covalent.
  e.g. TiC, $Mn_4N$, $Fe_3H$, $VH_{0.56}$ and $TiH_{1.7}$ etc.

$Fe_{0.96}O$



## Characteristics of Interstitial Compounds:

- ✓ High M.Pt (higher than those of pure metals)
- ✓ Very hard
- ✓ They retain metallic conductivity
- ✓ Chemically inert
  ↳ don't react

## Trends in the $M^{3+}/M^{2+}$ Standard Electrode Potentials

$E°(M^{3+}/M^{2+})$ : values show varying trend

→ **Low value for Sc** : shows that Sc in +3 has noble gas E.C.

→ **Highest value for Zn** : due to removal of an $e^-$ from stable $d^{10}$ configuration of $Zn^{2+}$

→ **High values for Mn** : shows that $Mn^{2+}(d^5)$: more stable

$Sc^{3+} : [Ar]^{18}$ ↳ $Sc^{2+}$ (X) $\mathcal{E}$: low

$Zn^{2+} \rightarrow Zn^{3+}$ (X) $\mathcal{E}$: high

$3d^{10}$ (Fully filled stable E.C)

$Mn^{2+} : 3d^5$ $\mathcal{E}$: high

↓ Half filled Stable E.C

# Trends in Stability of Higher Oxidation States:

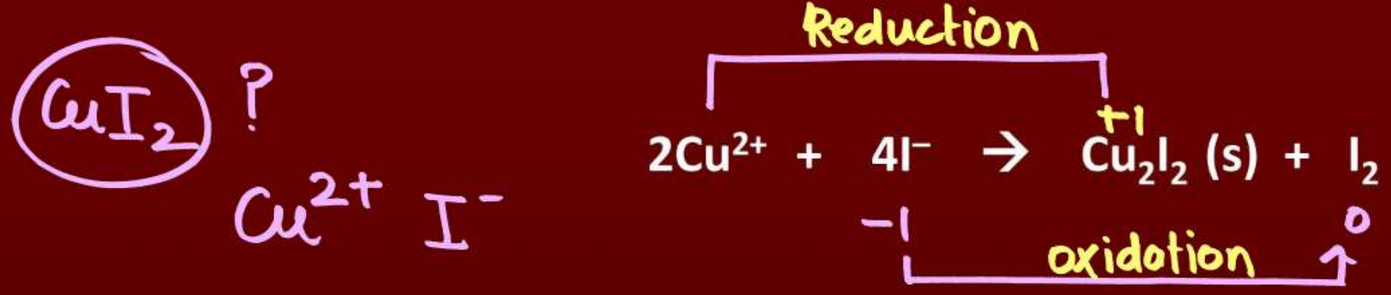
| Oxidation | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| +6 | | | $CrF_6$ | | | | | | |
| +5 | | $VF_5$ | $CrF_5$ | | | | | | |
| +4 | $TiX_4$ | $VX_4$[1] | $CrX_4$ | $MnF_4$ | | | | | |
| +3 | $TiX_3$ | $VX_3$ | $CrX_3$ | $MnF_3$ | $FeX_3$[1] | $CoF_3$ | | | |
| +2 | $TiX_2$[III] | $VX_2$ | $CrX_2$ | $MnX_2$ | $FeX_2$ | $CoX_2$ | $NiX_2$ | $CuX_2$[II] | $ZnX_2$ |
| +1 | | | | | | | | $CuX$[III] | |

**Formulae of Halides of 3d Metals**

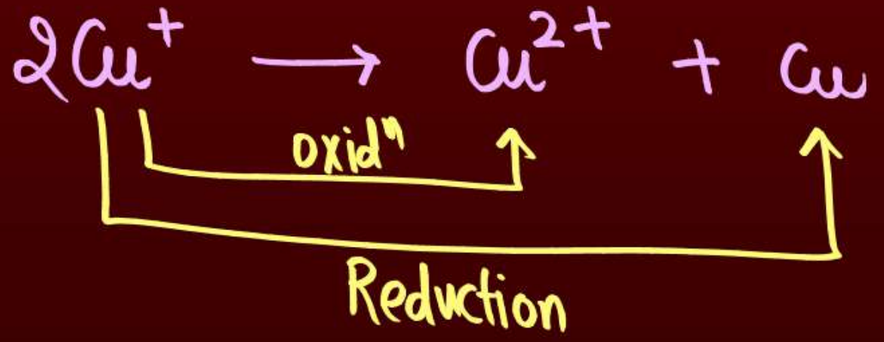
- ✓ The ability of fluorine to stabilize the highest oxidation state is due to either higher lattice energy (as in case of $CoF_3$) or higher bond enthalpy terms for the higher covalent compounds (e.g. $VF_5$ and $CrF_6$)
- ✓ Another feature of fluorides is their instability in the low oxidation states (e.g. $VX_2$, X = Cl, Br or I) and the same applies to CuX.

# Trends in Stability of Higher Oxidation States:

**All Cu(II) halides are known (except the iodide). In this case, $Cu^{2+}$ oxidises $I^-$ to $I_2$.**




However, many copper (I) compounds are unstable in aqueous solution & undergo disproportionation.

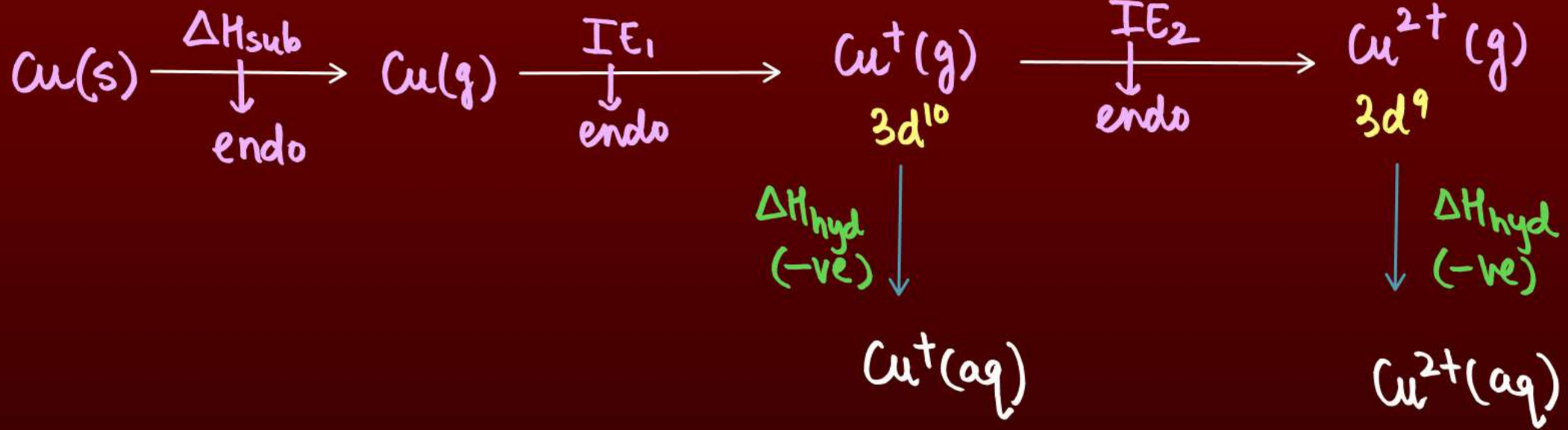


# Trends in Stability of Higher Oxidation States:

Note

$Cu^{2+}(aq) > Cu^{+}(aq)$

The stability of $Cu^{2+}(aq.)$ rather than $Cu^{+}(aq.)$ is due to the much more –ve $\Delta H^{\ominus}_{hyd}$ of $Cu^{2+}(aq.)$ than $Cu^{+}$, which more than compensates the second I.E. of Cu.



# QUESTION

Explain why $Cu^{+}$ ion is not stable in aqueous solutions?



# Trends in Stability of Higher Oxidation States:

Table 4.6: Oxides of 3d Metals

| Oxidation Number | Groups 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|
| + 7 | | | | | $Mn_2O_7$ | | | | | |
| + 6 | | | | $CrO_3$ | | | | | | |
| + 5 | | | $V_2O_5$ | | | | | | | |
| + 4 | | $TiO_2$ | $V_2O_4$ | $CrO_2$ | $MnO_2$ | | | | | |
| + 3 | $Sc_2O_3$ | $Ti_2O_3$ | $V_2O_3$ | $Cr_2O_3$ | $Mn_2O_3$ | $Fe_2O_3$ | | | | |
| | | | | | $Mn_3O_4$* | $Fe_3O_4$* | $Co_3O_4$* | | | |
| + 2 | | TiO | VO | (CrO) | MnO | FeO | CoO | NiO | CuO | ZnO |
| + 1 | | | | | | | | | $Cu_2O$ | |

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# Trends in Stability of Higher Oxidation States:

**Oxides of Chromium:** $\overset{+2}{CrO}$ $\overset{+3}{Cr_2O_3}$ $\overset{+4}{CrO_2}$ $\overset{+6}{CrO_3}$

→

As the oxidⁿ no. ↑ses, Acidic nature of oxides ↑ses

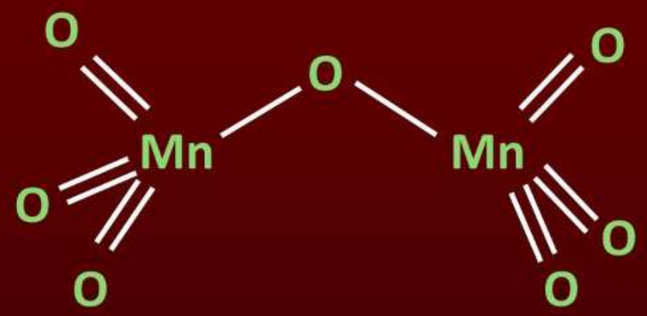
**Oxides of Manganese:** $\overset{+2}{MnO}$ $\overset{+3}{Mn_2O_3}$ $\overset{+4}{MnO_2}$ $\overset{+7}{Mn_2O_7}$

→

As the O.N ↑ses, Acidic character ↑ses

# Trends in Stability of Higher Oxidation States:

- The highest oxidation no. in the oxides coincides with the group no. and is attained in $Sc_2O_3$ to $Mn_2O_7$. The ability of oxygen to stabilize these high O.S. is more than fluorine.
  Thus; Highest fluoride of Mn is $MnF_4$ but oxide is $Mn_2O_7$.




The ability of oxygen to form multiple bonds to metals explains its superiority.

# QUESTION

$MnF_7$ is not stable while $Mn_2O_7$ is stable why?




Due to overcrowding of 7 Fluorine atoms around Mn



# QUESTION



**Gun metal is an alloy of :**

1. Cu and Al
2. Cu and Sn
3. Cu, Zn and Sn ✓
4. Cu, Zn and Ni (German Silver)

# QUESTION



Zinc and mercury do not show variable valency like d-block elements because

1. they are soft
2. their d-shells are complete
3. they have only two electrons in the outermost subshell.
4. their d-shells are incomplete

# QUESTION



The number of unpaired electrons in gaseous species of $Mn^{3+}$, $Cr^{3+}$ and $V^{3+}$ respectively are:

1. 4, 4 and 2
2. 3, 3 and 2
3. 4, 3 and 2
4. 3, 3 and 3

$_{25}Mn^{3+}: 3d^4$ ↑ ↑ ↑ ↑ _ 4 u.e

$_{24}Cr^{3+}: 3d^3$ ↑ ↑ ↑ _ _ 3 u.e

$_{23}V^{3+}: 3d^2$ ↑ ↑ _ _ _ 2 u.e

# QUESTION

The pair that has similar atomic radii is



$3d < 4d \approx 5d$

↳ Lanthanoid contraction

1. Mn and Re
2. Ti and Hf
3. Sc and Ni
4. Mo and W

| | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|
| 3d | 21 Sc | 22 Ti | 23 V | 24 Cr | 25 Mn | 26 Fe | 27 Co | 28 Ni | 29 Cu | 30 Zn |
| 4d | 39 Y | 40 Zr | 41 Nb | 42 Mo | 43 Tc | 44 Ru | 45 Rh | 46 Pd | 47 Ag | 48 Cd |
| 5d | 57 La | 72 Hf | 73 Ta | 74 W | 75 Re | 76 Os | 77 Ir | 78 Pt | 79 Au | 80 Hg |

# Homework

→ DPP Solve

→ NCERT Q's

# AI Board Booster: **Student se Super Student**

## Ek aisa course jo **AI ke use** se aapki Board prep level-up kare

Outcomes- Aapko kya milega? (padhai mein badlao)

**Faster revision:** Turn chapters into crisp **summaries, flashcards, and quizzes** in minutes.

**Target weak areas:** AI **flags your common mistakes** and suggests exact topics to revisit.

**Exam-style practice:** Convert **past papers into timed mocks** with stepwise solution.

**Showcase work:** Build **8–10 mini projects** (notes generator, quiz bot, image-to-notes, etc.).

**Repeatable study systems:** Unit tests & Boards ke liye **plug-and-play** checklists/templates.

**Responsible AI use:** Kya **allowed** hai, kaise cite karna hai, and **over-reliance** avoid karna

**Real-world edge: Prompting & automation** basics jo college/internships mein use karoge.

**Recognized proof: PW Skills certificate** *(signed by Founder Alakh Pandey).*

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### Course at a glance

- **Live Online | Hinglish | Mentor:** Vishal Tiwari Sir
- **18 sessions × 3 hrs (Saturdays)**
- **Weekly Doubt Room:** Mon, 45–60 min
- **Start:** 13 Sept 2025
- **Fee:** ₹2500 → ₹799 for Uday & Parishram paid users *(coupon in tray)*

**Limited seats - Offer ends soon**




# THANK YOU